

SIGNAL / RECURRENCE TEST SUMMARY

Goal:

I tested whether real-valued oscillatory input can be converted into controlled integer states, compared against output windows, and checked for repeatable recurrence/correlation behavior.

Important boundary:

These tests do NOT prove time prediction, universal determinism, or physical causality. They only test signal preservation, recurrence, and replay-style matching.

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TEST 1 — INTEGER PRE-DIGEST STABILITY

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Result:

PASS_INTEGER_PREDIGEST_STABILITY

Raw correlation: 0.879191

Integer correlation: 0.878066

Recovered correlation: 0.878066

Average input recovery error: 0.029243

Average output recovery error: 0.029122

Meaning:

Integer conversion preserved the oscillatory relationship with very little loss.

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TEST 2 — PHASE LOCK STABILITY

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Result:

PASS_PHASE_LOCK_STABILITY

Average window correlation: 0.856592

Phase-lock drift: 0.027710

Meaning:

The shifted input/output oscillation stayed stable across multiple windows.

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TEST 3 — MULTI-FREQUENCY RECURRENCE

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Initial result:

FAIL on direct recurrence.

Follow-up result:

PASS_INVERTED_MULTI_FREQUENCY_RECURRENCE

Average raw similarity: -0.566263

Average absolute similarity: 0.566263

Absolute similarity drift: 0.063854

Meaning:

The structure did recur, but mostly as an inverted/opposite-phase recurrence.

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TEST 4 — INTEGER SCALING STABILITY

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Result:

PASS_INTEGER_SCALING_STABILITY

x3 correlation: 0.992754

x9 correlation: 0.999253

x27 correlation: 0.999907

Best scale:

x27

Meaning:

Scaling into integer states preserved oscillatory structure extremely well.

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TEST 5 — NOISE STRESS TEST

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Result:

PASS_NOISE_ROBUSTNESS

Noise 0.05 correlation: 0.999248

Noise 0.10 correlation: 0.997374

Noise 0.20 correlation: 0.989089

Noise 0.35 correlation: 0.969935

Meaning:

Moderate noise did not destroy the recognizable oscillatory structure.

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TEST 6 — SYNTHETIC HISTORICAL REPLAY

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Result:

PARTIAL_HISTORICAL_REPLAY

Best historical match correlation:
0.885769

Follow-up inverted version:
PASS_INVERTED_HISTORICAL_REPLAY

Best absolute correlation:
0.933456

Meaning:
Synthetic oscillatory history contained replay-like recurrence windows, especially when inverted recurrence was allowed.

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TEST 7 — BASELINE COMPARISON

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Result:
PASS_REPLAY_BEATS_RANDOM_BASELINE

Best structured absolute correlation: 0.933456
Average random absolute correlation: 0.513693
Best random absolute correlation: 0.996044

Meaning:
Structured replay strongly beat average random matching, but one random window still beat it, so this is not prediction proof.

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TEST 8 — REPEATED HOLDOUT RECURRENCE

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Result:
PASS_REPEATED_HOLDOUT_RECURRENCE

25 holdouts tested.
Average score: 0.963449
Minimum score: 0.922331
Scores over 0.90: 25/25

Meaning:
Synthetic oscillatory history repeatedly produced strong prior recurrence matches.

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TEST 9 — REALISTIC NOISY SOLAR-LIKE REPLAY

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Result:

PASS_REALISTIC_REPLAY_RECURRENCE

25 holdouts tested.

Average score: 0.951024

Minimum score: 0.927435

Scores over 0.82: 25/25

Meaning:

A more realistic noisy/drifting solar-style synthetic signal still produced strong recurrence matches.

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TEST 10 — REAL MONTHLY SUNSPOT REPLAY

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Result:

PASS_REAL_WORLD_REPLAY_RECURRENCE

Real monthly sunspot samples loaded: 3328

Average score: 0.836584

Minimum score: 0.656776

Maximum score: 0.961079

Scores over 0.75: 16/20

Meaning:

Real historical sunspot data contains detectable replay-style recurrence windows.

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TEST 11 — REAL SUNSPOT PAST-TO-FUTURE SPLIT

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Result:

PASS_PAST_TO_FUTURE_REPLAY

Training samples: 2329

Future samples: 999

Average score: 0.848988

Minimum score: 0.694718

Maximum score: 0.957620

Scores over 0.72: 17/20

Meaning:

Earlier sunspot history contained window patterns similar to later unseen sunspot windows.

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CURRENT HONEST INTERPRETATION

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The grounded result is:

A deterministic replay/correlation framework can preserve and detect recurring oscillatory structures across synthetic signals, noisy synthetic signals, and real monthly sunspot data.

What this may be useful for:

- recurrence detection
- oscillatory signal analysis
- anomaly/state matching
- historical pattern replay
- solar-cycle similarity studies
- signal preprocessing experiments